AI

AI for Research

UTM BIG DATA CENTRE

MODULE 2 OF 5 · PRINTABLE NOTES

Idea Generation and Research Design

Generate and refine research ideas, explore potential research problems, formulate research questions and develop an initial research design with AI as a structured thinking partner.

FINER Criteria | Gemini Notebook | Research Question Design | Gap-Spotting

LEARNING OBJECTIVES

- Draft candidate research questions and an initial research design using a structured prompt.
- Assess any research question against the FINER criteria (Feasible, Interesting, Novel, Ethical, Relevant).
- Check the novelty of an idea using tools such as Scite, ResearchRabbit, Litmaps and Consensus.
- Use Gemini Notebook to compare uploaded sources and surface consensus, contradictions and gaps.
- Break a broad problem into a problem tree of researchable questions.
- Spot both explicit and implicit gaps across a small set of papers.

BEFORE YOU START

- Bring a broad topic or problem area you are already curious about. Even a rough one is enough to start.
- Have 2–3 papers ready to upload (from Module 1's shortlist) for the Gemini Notebook exercise.
- Create a free Google account if you do not already have one, so you can open Gemini Notebook during the session.

KEY CONCEPTS

Glossary of terms

Terms you will meet throughout this module, defined in plain language.

FINER
An acronym for Feasible, Interesting, Novel, Ethical and Relevant, used as a checklist for evaluating research questions.

Research gap
An unanswered question or unresolved contradiction in the existing literature.

Mixed-methods
A research design combining quantitative and qualitative data collection and analysis.

Source-grounded AI
An AI tool restricted to reasoning only over documents you explicitly provide.

Problem tree
A technique breaking one broad problem into distinct root causes, each yielding its own candidate research question.

Trainer's Tip, Dr. Shahizan: A good research question survives someone trying to break it. After Gemini Notebook suggests directions, ask a colleague or your supervisor to poke holes in the FINER assessment before you commit weeks of work to one direction.

AI TOOLS FOR THIS MODULE

AI tools used in Module 2

TOOL	CATEGORY	WHAT IT DOES	WHERE TO ACCESS
Gemini Notebook	Source-grounded AI	Builds answers only from documents you upload — ideal for comparing a shortlist of papers.	notebooklm.google.com
Scite	Citation context	Shows whether citations support, contrast with, or merely mention a publication.	scite.ai
ResearchRabbit	Research landscape	Maps related papers, authors and citation connections for iterative exploration.	researchrabbit.ai
Litmaps	Literature mapping	Creates citation maps and monitors new publications on a topic.	litmaps.com
Consensus	Evidence search	Synthesises findings across papers and links every answer back to its source.	consensus.app

HOW TO USE

Step-by-step tool guides

STEP-BY-STEP GUIDE

Gemini Notebook

Gemini Notebook builds answers only from documents you upload. Upload the handful of papers behind your idea, then ask it to compare them — a fast way to surface contradictions, gaps and unanswered questions before you commit to a research question.

01

Upload

Upload 3 to 5 key papers on your topic, either your own shortlist or highly-cited papers found with Scopus AI or Publish or Perish in Module 1.

02

Ask

Ask Gemini Notebook to compare the papers and surface consensus, contradictions and unanswered questions, using the gap-finding prompt below.

03

Refine

Use what surfaces to sharpen your candidate research question and confirm it addresses a genuine gap.

GAP-FINDING PROMPT

Compare the uploaded sources on [TOPIC]. Identify: (1) areas of consensus, (2) contradictions or unresolved debates, (3) one specific gap that has not yet been addressed. Use only the uploaded sources, and flag anything uncertain or not supported.

HANDS-ON PRACTICE

All exercises, with model answers

Every exercise below matches the hands-on activity on the module's web page. Attempt each one yourself first, then compare against the model answer.

EXERCISE 2.1

~10 min

Research Question & Design Prompt Builder

Describe your topic and constraints. The builder assembles a ready-to-use prompt asking an AI chat tool to propose candidate research questions, assess them against the FINER criteria, and outline an initial research design.

- [ ] State your broad topic or interest area in one sentence.
- [ ] Note your field/discipline, target population or context, and the motivation or observed problem behind the topic.
- [ ] Choose a preferred research type (quantitative, qualitative, mixed-methods, systematic review, or not yet decided) and any known constraints.
- [ ] Run the assembled prompt in an AI chat tool, then check every candidate question against FINER yourself before choosing one.

✓ WORKED EXAMPLE — GENERATED PROMPT

Sample input — Topic: "use of generative AI among postgraduate researchers" · Field: Educational Technology · Context: Malaysian public university postgraduates · Type: Mixed-methods.

Generated prompt: "Act as an experienced research supervisor in Educational Technology. I am exploring this broad topic or interest area: use of generative AI among postgraduate researchers. Target population or context: Malaysian public university postgraduates. Preferred research type: Mixed-methods. Propose 3 to 5 candidate research questions for this topic. For each question, assess it against the FINER criteria (Feasible, Interesting, Novel, Ethical, Relevant) in one line each. Then outline an initial research design (suggested methodology, data source and analysis approach) for the strongest question. Important: do not fabricate literature, citations or findings. Clearly flag anything I should verify against the published literature before committing to a direction."

Treat the AI's response as a draft to interrogate, not a final answer.

EXERCISE 2.2

~15 min

Pressure-test your idea with Gemini Notebook

Gemini Notebook builds answers only from documents you upload. Upload the handful of papers behind your idea, then ask it to compare them. This is a fast way to surface contradictions, gaps and unanswered questions before you commit to a research question.

- [ ] Upload 3 to 5 papers into a new Gemini Notebook, either your own shortlist or results from Module 1's Scopus AI or Publish or Perish exercise.
- [ ] Paste the gap-finding prompt from the tool guide above and review what Gemini Notebook surfaces.
- [ ] Note one contradiction or gap you did not expect, and consider whether it sharpens your candidate research question.
- [ ] Cross-check any claimed gap against a proper database search (Module 1) before treating it as genuinely novel — Gemini Notebook only reasons over what you upload.

✓ **MODEL ANSWER — SAMPLE OUTPUT**

- Topic uploaded: 4 papers on AI-assisted feedback in higher-education writing courses (2020–2023).
- Consensus found: all 4 papers agree AI feedback reduces the time students spend on first-draft revision.
- Contradiction found: two papers report higher student satisfaction with AI feedback, while one reports no significant difference — flagged as an unresolved debate.
- Specific gap identified: none of the 4 papers examine whether the effect holds for postgraduate (as opposed to undergraduate) writers.
- Verification step taken: ran a quick Scopus search for "postgraduate AI feedback writing" to confirm this gap was not simply missed by the 4 uploaded papers — found only 1 loosely related hit, supporting that the gap is likely genuine.

EXERCISE 2.3

~15 min

Problem tree: from broad problem to candidate questions

A problem tree breaks one broad problem into its root causes, and each root cause can become its own candidate research question. It stops you jumping straight to a vague RQ that is really three different questions stacked together.

- [ ] Write your broad problem or interest area at the top (e.g., "students struggle to use AI tools responsibly in coursework").
- [ ] List 3 to 4 distinct root causes that could each explain part of the problem.
- [ ] For each root cause, draft one candidate research question that addresses just that cause.
- [ ] Check each candidate question against FINER before you pick one to pursue further.

✓ **WORKED EXAMPLE**

- Problem: Students struggle to use AI tools responsibly in coursework.
- Cause 1 — unclear institutional policy → RQ: How do postgraduate students interpret their university's AI use policy?
- Cause 2 — no verification habit → RQ: What verification practices, if any, do students apply to AI-generated content?
- Cause 3 — lack of training → RQ: Does a short AI-literacy workshop change students' verification behaviour?

EXERCISE 2.4

~15 min

Gap-spotting worksheet

Papers often state a gap directly, usually near the end, and also leave gaps unstated that you can only spot by comparing several papers together. This exercise practises finding both.

- [ ] Take 3 to 5 papers from your Module 1 shortlist or literature review matrix.
- [ ] For each paper, search the discussion or conclusion for an explicit gap statement (phrases like "future research should" or "this study did not examine").
- [ ] Then look across all the papers together for an implicit gap, something none of them addressed, or a contradiction between two papers.
- [ ] Record both types of gap, and mark which ones look genuinely novel versus already being addressed elsewhere.

SOURCE	EXPLICIT GAP STATEMENT	IMPLICIT GAP (ACROSS SOURCES)

MODEL ANSWER — WORKED EXAMPLE

- Explicit gap: "Future research should test this framework in non-Western contexts" (Smith, 2021) — directly usable; quote it and cite the paper as the source of the gap.
- Implicit gap: if three papers all studied undergraduate samples and none studied postgraduates, that is a gap you have to spot yourself by comparing sources, not one any single paper states outright.
- Genuinely novel vs already addressed: search for the gap topic directly (e.g., "AI literacy postgraduate non-Western") before assuming it is unaddressed — a gap may already have been closed by a very recent paper your search missed.

EXERCISE 2.5

~10 min

Peer critique with the FINER rubric

Your own research question is hard to critique objectively. Trading questions with a partner and scoring them against FINER surfaces weaknesses you would otherwise miss.

- [ ] Pair up with another participant and exchange your candidate research questions.
- [ ] Score your partner's question from 1 (weak) to 5 (strong) on each FINER criterion.
- [ ] For any score of 3 or below, write one specific reason and one suggestion to strengthen it.
- [ ] Swap back, discuss the feedback, and revise your question if needed.

FINER CRITERION	SCORE (1–5)	COMMENT / SUGGESTION
Feasible		
Interesting		
Novel		
Ethical		
Relevant		

✓ MODEL ANSWER — SAMPLE COMPLETED CRITIQUE

Sample RQ critiqued: "How does AI feedback affect student writing?"

Feasible — 4/5, doable within a semester with an existing class.
Interesting — 3/5, relevant but not sharply framed; suggestion: name the specific outcome (revision quality, satisfaction, or grades).
Novel — 2/5, heavily studied at undergraduate level; suggestion: narrow to postgraduate writers or a specific discipline to sharpen novelty.
Ethical — 5/5, low risk, standard consent process suffices.
Relevant — 4/5, clearly useful to writing-centre practice.

Revised question after feedback: "How does AI feedback affect the revision quality of postgraduate students' thesis chapter drafts?"

WHY USE AI HERE

Traditional vs AI-assisted approach

Task	Traditional Approach	AI-Assisted Approach
Drafting candidate questions	Brainstorm alone or wait for a supervisor meeting	RQ Builder proposes 3–5 candidates in minutes, ready to discuss
Checking against FINER	Manually reason through each criterion for each question	The generated prompt scores every question against FINER automatically
Testing novelty	Search separately and judge by memory of what you have read	Scite or Consensus surface support, contradiction or consensus instantly
Stress-testing against sources	Re-read every paper to compare them against each other	Gemini Notebook compares uploaded sources and surfaces contradictions directly

The trade-off: AI accelerates the first draft of an idea; you still confirm it addresses a genuine, ethical and feasible gap.

COMMON PITFALLS

Things to verify before you trust the output

- [ ] Treating an AI-proposed research question as final rather than as a draft to interrogate against FINER yourself.
- [ ] Assuming a gap Gemini Notebook surfaces is genuinely novel without cross-checking it against a proper database search — the tool only reasons over what you upload.
- [ ] Merging multiple root causes into one vague research question instead of using the problem tree to separate them.
- [ ] Skipping the peer-critique step because your own question "feels" strong — self-assessment misses weaknesses a partner catches immediately.
- [ ] Confusing an explicit gap (stated by the paper) with an implicit gap (visible only across several papers) and citing the wrong kind of evidence for each.

Key Takeaways

- A strong research question is Feasible, Interesting, Novel, Ethical and Relevant (FINER).
- AI can propose and assess candidate questions quickly, but you decide which one to pursue.
- Check novelty with tools like Scite, Consensus or Litmaps before assuming a gap is real.
- Gemini Notebook only reasons over what you upload, and it cannot see the wider literature.
- A problem tree and gap-spotting worksheet both stop a vague idea from turning into a vague research question.

> **What's next:** Have a candidate research question? Module 3 helps you organise the literature behind it into a structured mind map or synthesis matrix.

FOR YOUR NOTES

Session Notes

FOR YOUR NOTES

Session Notes (continued)